

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)
Course Name: Applied Mathematics-I Course Code: MA-111 Examination - 1st periodical
Scheme - CBCS Branch: CSE(AI/ML) Sem: 1st
Question Paper for 1st Periodical Examinations (October, 2023)

Time: 2 Hours

Max. Marks: 50

Note: All questions are compulsory.

Course Outcomes:

On successful completion of this course the students should be able to:

CO1. Analyze the convergence/divergence of sequences and series along with error estimates.

CO2. Explain and apply Mean value theorems, Maclaurin's series, Taylor series and their applications.

CO3. Apply integral calculus to evaluate definite and improper integrals, length of arc, surface areas and volume of revolution.

CO4. Identify limits, continuity and partial derivatives of function of several variables.

CO5. Evaluate the maxima and minima by using higher order partial derivatives and Lagrange's method of multipliers.

CO6. Illustrate multiple integrals in different co-ordinates along with their applications in Engineering

Q.1 (a) Define convergent sequence? Examine the convergence of the sequence $a_n = \sin n$. (CO1)

(b) Define Oscillatory series? When the series $\sum (2x)^n$, for $n = 0$ to ∞ , is said to be convergent? (CO1)

(d) Find the value of c of the Lagrange's mean value theorem if $f(x) = x(x - 1)(x - 2)$ in $[0, 1/2]$. (CO2)

(e) Define arc length of a plane curve? Write the formula for area of surface of revolution if x -axis is the axis of rotation? (CO3)

(f) What do you mean by Gamma function? Write the relation between Beta and Gamma function.

(CO3)

(5×2=10)

Q.2 (a) Examine the convergence of the following series $\sum (1/n) \sin(1/n)$ (CO1) (5)

(b) State and prove Rolle's mean value theorem. (CO2) (5)

Q.3 (a) Check whether the following series is convergent/divergent, absolutely/conditionally convergent? (CO1) (5)

$$5/2 - 7/4 + 9/6 - 11/8 + \dots \infty$$

(b) Find the radius of convergence of power series $\sum (1 + 1/n)^{n^2} z^n$. (CO1) (5)

Q.4 Find the arc length of the astroid $x^{2/3} + y^{2/3} = a^{2/3}$. (CO3) (10)

Q.5 Find the volume of the solid region bounded above and below by paraboloids $x^2 + y^2 + z = 6$ and $2x^2 + y^2 = z$, respectively. (CO3) (10)

(Dr. Kiran Thakur)